Analytic Chemistry
deals with detection of substances present in sample (qualitative analysis and measurement of their amount (quantitative analysis)

Qualitative analysis for Inorganic salt

Acid + Base ⟶ salt + H2O (Normalization test)

HCl + NaOH ⟶ NaCl + H2O

NaCl → $Na^{\oplus}$ (Basic radical, Positively Charged Ion, Cation) + $Cl^{\ominus}$ (Acidic radical, negatively Charged Ion, anion)

Qualitative analysis for Acidic Radical

I	II	III
CO_3^{-2}	Cl^-	
HCO_3^-	Br^-	SO_4^{-2}
S^{-2}	I^-	PO_4^{-3}
SO_3^{-2}	NO_3^-	$B_4O_7^{-2}$
$S_2O_3^{-2}$		
NO_2^{-2}		
HCl (dil) (diluted)	H_2SO_4 conc (Concentrated)	Barle form PPt (Precipitate)

I: evolves gases → Colour, odour

II: evolves gases → Color

III: form PPt

Physical Properties

In Solid

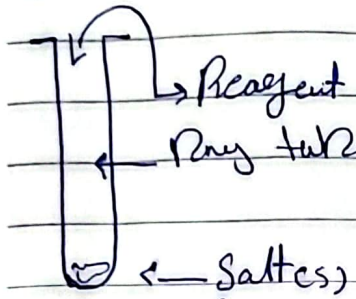
Colour : white, yellow ---, colourless

Shape : Powder, Fine Crystal, Crystal, sheet

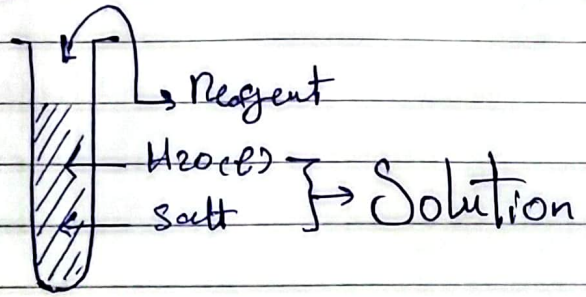
Solibility : Soluble or Insoluble in H_2O

Chemical test

① main test (Dry test)



② Confirmal test (wet test)



Experment	observation	results	
$D_{salt} + HCl_{(aq)}$	① effervesence and evolution CO_2 gas which turbied $Ca(OH)_2$ water	A.R may be CO_3^{-2}, HCO_3^-	obs ③ which turn Paper of lead acetate into Black
	② reddish brown fumies at mouth of test tube	A.R may be NO_2^-	
	③ evolution of H_2S gas with rotten odour	A.R may be S^{-2}	

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Observation	Results
④ Evolution of SO_2 gas and yellow ppt formed by shaking and standing	A.R may be $S_2O_3^{-2}$
⑤ Evolution of gas which turn $K_2Cr_2O_7$ into green from orange	A.R may be SO_3^{-2}
⑥ negative -ve	SO_3^{-2} is absent A.R may be SO_4^{-2} or Cl^-

Confirmal test

[1] for CO_3^{-2}, HCO_3^-

Exp	CO_3^{-2}	HCO_3^-
① Salt sol. + $AgNO_3$ + $BaCl_2$ + $MgSO_4$ + $HgCl_2$	white ppt on cold	white ppt after heating
	red-brick brown on cold	red-brick brown after heating

[2] for S^{-2}

Exp	obs	my
① Salt sol. + $AgNO_3$ or $Pb(CH_3COO)_2$	black ppt	A.R is S^{-2}

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[3] For SO_3^{-2}

Exp	Obs	res
① Salt soln + $AgNO_3$ + $Pb(CH_3COO)_2$	white ppt	A.R is SO_3^{-2}
② Salt soln + I_2	brown color of I_2 disappear	$I_2 \rightarrow I^-$

[4] For $S_2O_3^{-2}$

Exp	Obs	res
① Salt soln + $AgNO_3$ + $Pb(CH_3COO)_2$	white ppt turn to black by heating	A.R is $S_2O_3^{-2}$
② Salt soln + I_2	brown color of I_2 disappear	

[5] For NO_2^-

Exp	Obs	res
① Salt soln + KMnO4 + H_2SO_4 (dil)	Color of Pr Permanganate disappeared	A.R is NO_2^-
② Salt soln + KI + H_2SO_4 (dil)	give brown Color due to formation of I_2	$I^- \rightarrow I_2$

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